

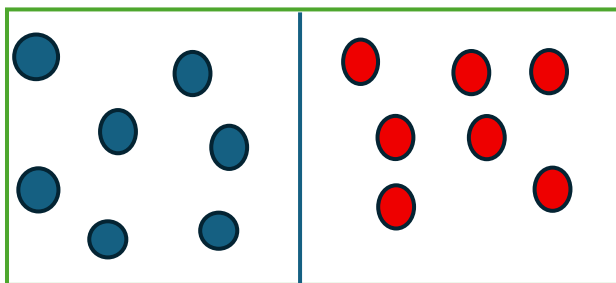
ENTROPY hi eng nge a nih chiah le?

Science zir tawh hrim hrim chuan ‘**Entropy**’ hi kan hre theuh tawhin a rinawm. Entropy hi thil interesting deuh mai hi a ni a. A definition han in zawt ta ila, mi tam zawk rilru ah chuan textbook/ google atanga kan hriat heng buzzwords: **chaos**, **disorder** etc tih hi kan rilru a lo lang hmasa ber pawl an ni maithei. Class 11 textbook a entropy kan define dan chu “**It is the measure of disorder of a system**” kan ti mai thin a. Mahse uluk zawka ngaihtuah hian he definition hian ngaihtuah chhunzawm a dawl lo hle. Entir nan ‘disorder’ tih chu eng chiah nge, lei vut pahnih hi vawm ta ila, engtin nge entrophy chu kan teh theih ang? Entropy concept hriatthiamna atan min pui thuilo hle a. Hei aia definition tha hi a awm thei em?

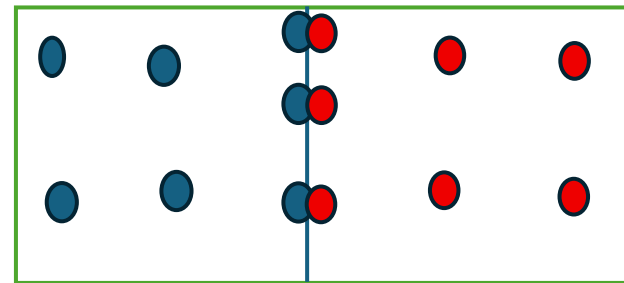
A tlangpuiin ‘**Entropy**’ hi Physics, Chemistry leh Engineering ah chuan **thermodynamics** atanga kan hmelhriat a ni tlangpui a. Tin Computer science lamah **information entropy** tih concept hran te an nei ve leh thung a. Vawiin ah chuan thermodynamics lam atanga entropy kan thlir dan lam kan sawi anga.

Entropy kan sawi hma in example atan thil pahnih ka rawn hmang anga chumi te atang chuan hriatthiam kan tum dawn a ni.

System pakhat ka nei a, chutah chuan atoms blue colour leh atoms red colour awm ta se, an inkarah indanna awm ta se, a indanna hi hun engemaw chen hnuah la bo ta ila, a thleng tur chu kan hre vek awm e. A lo in mix dawn a ni tiraw?



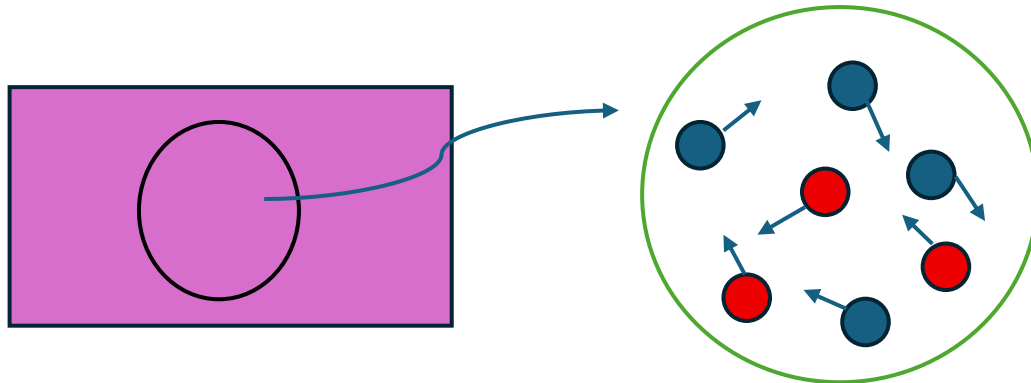
At $t=0$



At $t=t_1$



At $t \gg t=0$

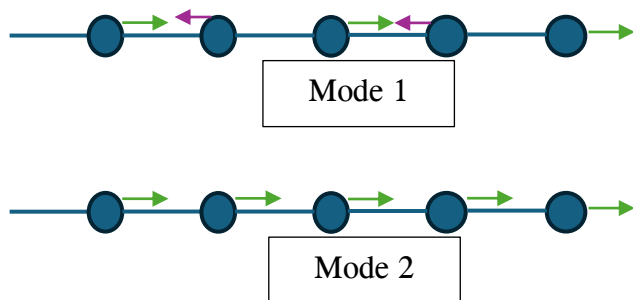


Kan han zoom in tak tak chuan atom blue leh red te a hrana la awm, han insu kual vel kan la hmu thei dawn a ni. Mahse khimi mixture khi macroscopic (mit in a hmuh theih thil) ah chuan a tira red leh blue khan kan $\dot{\text{t}}\text{endarh}$ thei miahlo thung. Khimi atoms kan zoom in ho erawh khi chu han insu vel mah se a tira an awm dan (blue and red awm hrang) ah khan an let leh thei tlat a ni. **A tawz zawng chuan small scale (microscopical level, length scale te) ah chuan he mixing process hi a reversible (a let zawng in a kal thei a), mahse heng zawng zawng lak khawm mit a hmuh theih (macroscopical level, length scale lian) hi chu a reversible ve leh miahlo thung si a ni!**

Chuvang chuan issue pakhat kan nei ta phawt: Eng vengin nge irreversible process te kha microscopic level ah reversible process an nih si? **“Length scale issue”** kan nei ta phawt a. A issue dang leh chu **“Time scale issue”** a ni a, hetiang hian sawi fiah ta ila:

Kan hmanraw $\dot{\text{t}}$ ha ber hmangin, molecules te in su lai hi en tum ta ila. Kan camera speed kha a tam berah 10000 frames per second pawh nise molecules te insut na speed (collision speed) hi room temperature ah 10^{-10} seconds vel a ni a chuvang chuan kan camera frame 1 leh 2 inkarah collision million 1 vel a lo thleng hman dawn tihna a nih chu!

Kan hmanraw $\dot{\text{t}}$ ha ber hmanga kan teh theihloh thil kha a awm miahlo tihna a nilo a, kan zuk observe theilo mai a ni. Example pakhat hmang in han sawi fiah leh dawn ta ila:



Hemi system ah hian heng atoms 5 te in pawh dan hi lo en ta ila. Tarmit special em em nei ta bawh ila.

A chung zawk mode 1 hi chu pawt let tu (opposing forces) an awm avangin a kan special tarmit hmang in a khawma a che dul dul lai kan hmu thei mang dawnlo a ni. Mahse a hnuai

zawh mode 2 zawh hi en ta ila, an chet la lam a inan vek avangin kan special tarmit hmang in observe a awl zawh dawn a ni. Mahse zawhna inzawt tur che a ka duh tak chu: Kan special tarmit hmanga kan observe ve theihloh mode khi a awm ve lo tihna a ni em? Ni lo e. Kan observe theihloh avang khan a awm ve lo tihna a lo ni miahlo.

Hemi information kan neih kimloh na hi Entropy in min quantify sak tak chu a ni! Entropy chu eng nge a nih kan tih chuan: Kan system chungchang kan hriat kimloh na min hrih fiah tu a ni tihna a nih chu!

Kan chik tur thil pakhat chu, entropy kan explain nan hian thermodynamics a kan zir dan temperature lam engmah rin lut tel lo in kan hrih fiah zel hrih a ni tih hi. Hei hi thil ho te a ni miahlo! Disorder kan tih ringawt khan entropy hi a sawi fiah zo in ka hrelo fo thin a. He entropy definition 'information' lam ang zawnga kan rinluh hi ka ngaihdan chuan a **satisfying** hle in ka hria a ni.

Anih chuan engtin nge Entropy chu kan quantify tak ang?

Entropy chungchang postulate 3 ka lo zia anga, chuta tang chuan eng function in nge satisfy thei tih kan en ho dawn a ni.

Postulate number 1: If there are $m > n$ possible microstates for any macrostate then

$$s(m) > s(n)$$

Hemi postulate in a sawi awmzia ber chu microstate m leh microstate n kha awm ta se, microstate m khan in rem theih dan nghah zawh ta se, a entropy kha a sang zawh dawn tihna a ni. In rem theih dan combination a tam paih leh kan 'LACK OF INFORMATION' khami system tan khan a tam telh telh a. Chu chuan entropy a ti sang tihna ti ila kan sawi sual awm lo e.

Postulate number 2: If there is only one microstate then $s(1)=0$.

Hemi postulate in a sawi awmzia chu, i microstate kha chi khat chiah in rem theih ta se. Entirnan i bag ah marble 1 chiah leng ta se, i marble kha i bag ah chi dang zawngin rem dan dang a awm miahlo tihna a ni a, chuvang chuan 'LACK OF INFORMATION' kha a awm miahlo tihna a ni. Chuvang chuan entropy kha zero a ni daigh dawn tihna a nih chu.

Postulate number 3: Take mn states, $s(mn)=s(m)+s(n)$

Hemi postulate in a sawi awmzia chu state hran pahnih kha combine ta la, an entropy kha a multiplicable lo a, a additive a ni. Multiplicable ta se additive aiah postulate number 2 khi a break nghal daih dawn tihna a ni (Lo try ve chhin teh).

Anih chuan heng postulate 3 te hi eng function in nge satisfy thei ta ang? Tih kha kan zawhna dawt leh chu a ni a. Entropy hi a mathematical equation kan ziah chhuah a ngai si a.

$$\ln(x) = \frac{p}{q} \quad (\text{both are integers})$$

$$q \ln(x) = p$$

$$\ln(x)^q = p$$

$$x^q = e^p$$

$$s(x^q) = se^p$$

$$s(x) + s(x^{q-1}) = s(e) + s(e^{p-1})$$

Principle of mathematical induction hmangin:

$$q s(x) = p s(e)$$

$$s(x) = \frac{p}{q} s(e)$$

$$s(x) = K \ln x$$

A tawp bera equation kan neih hi kan hriat lar em em **Boltzmann relation** kha a ni a, he relation hian entropy ‘lack of information’ kha mathematical expression a rawn pe a, a rawn quantify ta a ni.

Kan sawi tak ang khan entropy chu **kan system chungchang kan hriat kimloh na tehna a ni.**

Entir nan,

Particle pahnih nei ta ila, and energy belh khawm kha 10 Joules ni ta se. Chumi particles ho entropy chu kan chhut thei chiah ta a ni. Particle 2 hi kan microstates an ni a, value hrang hrang an nei thei a ni, mahse an value belh khawm hi 10 J a nih a ngai a ni. (10J hi kan macroscopic measurement angah ngai ila). Hemi example a kan hriat loh chu, microstate ho kha engtin nge an in belh theih hi a ni. Entir nan particle te khan khan energy {1,9}{2,8}{3,7}{4,6}{5,5} an nei tih kan hre lo. Awm theih dan 5 an nei. Chuvangin entropy chu $\log(5)$ a ni dawn tihna a ni. Entropy chu kan teh thei ta!